

DSC5003: STORING AND RETRIEVING DATA

Effective Term

Semester A 2026/27

Part I Course Overview

Course Title

Storing and Retrieving Data

Subject Code

DSC - Data Science

Course Number

5003

Academic Unit

Data Science (DS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

SDSC5003 Storing and Retrieving Data

Exclusive Courses

Nil

Part II Course Details

Abstract

This course offers knowledge of the relational database and an introduction to Hadoop/Spark system including the entity-relationship model for designing the relational database, principles of the database development process, the Structural Query Language for retrieving and storing data via a database, and the introductory level Hadoop/Spark content.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Describe principles of the relational database design and structural query language syntax	30	x		
2	Explain the efficiency issue in database systems, including storage and indexing as well as query optimization	30	x	x	
3	Discuss modern database techniques such as JSON and NoSQL	10	x	x	
4	Explain the MapReduce computing framework and the basics of Spark	10	x	x	
5	Implement taught knowledge to develop a database application	20		x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Students will engage in lectures introducing relational database systems, principles of database design and development, syntax of structural query language, as well as fundamentals and introductory applications of Hadoop/Spark	1, 2, 3, 4	26 hours/semester

2	Laboratory work	Students will engage in developing the ability of designing and developing the relational and big database as well as differences between retrieving and storing data via relational and big databases to generate applications.	2, 3, 4, 5	13 hours/semester
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Additional Information for LTAs

Lectures cover not only the narrowly focused techniques in engineering economy but also the wider issues of the environment that affect engineering economic decision making. Students are expected to participate in class discussion when needed.

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Group Project	2, 3, 4, 5	35	While it' s acceptable to seek help from AI tools such as ChatGPT during a group project, avoid outsourcing your work completely. The key is to ensure you fully comprehend the content produced by the AI.	Yes
2	Individual Coursework	1, 2, 3, 4	15	It' s okay to use AI tools like ChatGPT for help with your assignments— but don' t let AI replace you. Instead, let it assist you. The key is to stay actively involved, critically evaluate the output, and ensure you truly understand the content generated.	Yes

Continuous Assessment (%)

50

Examination (%)

50

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark for the examination should be obtained.

Assessment Rubrics (AR)

Assessment Task

Group Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

35%

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Individual Coursework (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

15%

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

50%

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Group Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

35%

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Individual Coursework (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

15%

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

50%

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Part III Other Information**Keyword Syllabus**

- Introduction of Database and Its Development Process
- Data Modeling (Entity-Relationship model, meta model)
- Database Design Process and development
- Structured Query Language in Relational Database
- Storage and Indexing
- Query Optimization
- JSON and NoSQL
- Fundamentals of MapReduce and Spark
- Applications of Hadoop/Spark

Reading List**Compulsory Readings**

Title	
1	Lecture Notes

Additional Readings

Title	
1	NIL